United States Patent	12407596
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Schunter; Matthias

Technologies for protocol execution with aggregation and caching

Abstract

Technologies for protocol execution include a command device to broadcast a protocol message to a plurality of computing devices and receive an aggregated status message from an aggregation system. The aggregated status message identifies a success or failure of execution of instructions corresponding with the protocol message by the plurality of computing devices such that each computing device of the plurality of computing devices that failed is uniquely identified and the success of remaining computing devices is aggregated into a single success identifier.

Inventors:	**Schunter; Matthias (Seeheim-Jugenheim, DE)**
Applicant:	**Intel Corporation** (Santa Clara, CA)
Family ID:	**56130777**
Assignee:	**INTEL CORPORATION (Santa Clara, CA)**
Appl. No.:	**18/459724**
Filed:	**September 01, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230412484 A1	Dec. 21, 2023

Related U.S. Application Data

continuation parent-doc US 17445163 20210816 US 11750492 child-doc US 18459724
continuation parent-doc US 16863169 20200430 US 11121958 20210914 child-doc US 17445163
continuation parent-doc US 15860301 20180102 US 10644984 20200505 child-doc US 16863169
continuation parent-doc US 14580758 20141223 US 9860153 20180102 child-doc US 15860301

Publication Classification

Int. Cl.: **H04L12/26** (20060101); **H04L12/24** (20060101); **H04L29/08** (20060101); **H04L41/0686** (20220101); **H04L43/18** (20220101); **H04L67/566** (20220101); **H04L67/568** (20220101)

U.S. Cl.:

CPC **H04L43/18** (20130101); **H04L41/0686** (20130101); **H04L67/566** (20220501); **H04L67/568** (20220501);

Field of Classification Search

CPC: H04L (43/18); H04L (67/566); H04L (67/568); H04L (41/0686); H04L (67/1091); H04L (12/26); H04L (29/08); H04L (12/24)

USPC: 709/224

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5924116	12/1998	Aggarwal et al.	N/A	N/A
6161208	12/1999	Dutton et al.	N/A	N/A
6216199	12/2000	DeKoning et al.	N/A	N/A
6256747	12/2000	Inohara et al.	N/A	N/A
6629199	12/2002	Vishlitzky et al.	N/A	N/A
7386608	12/2007	Tsai et al.	N/A	N/A
7603458	12/2008	Sexton et al.	N/A	N/A
7769885	12/2009	Kompella	N/A	N/A
8332502	12/2011	Neuhaus et al.	N/A	N/A
8539109	12/2012	Lee et al.	N/A	N/A
9258335	12/2015	Taylor	N/A	N/A
9860153	12/2017	Schunter	N/A	H04L 67/568
10644984	12/2019	Schunter	N/A	H04L 41/0686
11121958	12/2020	Schunter	N/A	H04L 67/568
11750492	12/2022	Schunter	N/A	H04L 67/566
2003/0140112	12/2002	Ramachandran et al.	N/A	N/A
2005/0129183	12/2004	Booth	N/A	N/A
2006/0173985	12/2005	Moore	N/A	N/A
2006/0265489	12/2005	Moore	N/A	N/A
2007/0073894	12/2006	Erickson	707/E17.108	H04L 67/51
2007/0079062	12/2006	Miyawaki et al.	N/A	N/A
2007/0133556	12/2006	Ding et al.	N/A	N/A
2007/0143549	12/2006	Saha et al.	N/A	N/A
2007/0174419	12/2006	O'Connell	709/217	G06F 11/0781
2009/0013210	12/2008	McIntosh et al.	N/A	N/A

2009/0027061	12/2008	Curt et al.	N/A	N/A
2009/0146822	12/2008	Soliman	705/3	A61B 5/0002
2010/0049815	12/2009	Vanecek et al.	N/A	N/A
2011/0029461	12/2010	Hardin, Jr.	N/A	N/A
2011/0182189	12/2010	Martin	N/A	N/A
2013/0104251	12/2012	Moore et al.	N/A	N/A
2013/0297708	12/2012	Beerse et al.	N/A	N/A
2013/0339473	12/2012	McCaffrey et al.	N/A	N/A
2014/0052922	12/2013	Moyer et al.	N/A	N/A
2014/0053003	12/2013	Moyer et al.	N/A	N/A
2014/0089591	12/2013	Moir et al.	N/A	N/A
2014/0181247	12/2013	Hindawi et al.	N/A	N/A
2014/0219021	12/2013	Trantham et al.	N/A	N/A
2014/0351713	12/2013	Hallerström Sjöstedt et al.	N/A	N/A
2015/0006622	12/2014	Lim et al.	N/A	N/A
2015/0019788	12/2014	Adler et al.	N/A	N/A
2015/0156194	12/2014	Modi et al.	N/A	N/A
2015/0208286	12/2014	Ozturk	370/331	H04W 36/0022
2016/0036771	12/2015	Yadav et al.	N/A	N/A
2016/0072667	12/2015	Zhu et al.	N/A	N/A
2016/0157221	12/2015	Kim	N/A	N/A
2016/0182343	12/2015	Schunter	N/A	N/A
2018/0145898	12/2017	Schunter	N/A	N/A
2021/0044549	12/2020	Renjith	N/A	N/A

OTHER PUBLICATIONS

Mark Manulis and Jorg Schwenk, 2009, “Security model and framework for information aggregation in sensor networks”, ACM Trans. Sen. Netw. 5.2, Article 13, (Apr. 2009), 28 pages,. DOI=10.1145/1498915.149819 http://doi.acm.org/10.1145/1498915.1498919. cited by applicant

Hani Alzaid, Ernest Foo and Juan Gonzalez Nieto, 2008, “Secure Data Aggregation in Wireless Sensor Network: A Survey”, In proceedings of the sixth Australian Conference on Information Securit—vol. 81 (AISC '08), Ljiljana Brankovic and Mirka Miller (Eds.), vol. 81, Australian Computer Society, Inc., Darlinghurst, Australia, 93-105. cited by applicant

Boneh, D., Gentry, C., Waters, B., “Collusion Resistant Broadcast Encryption with Short Ciphertexts and Private Keys”, Shoup, V. (ed) Crypto'05. LNCS, vol. 3621, pp. 258 (275. Springer, Berlin, Heidelberg, (2005). cited by applicant

Non-Final Office Action in U.S. Appl. No. 18/356,587, mailed Apr. 25, 2024, 7 pages. cited by applicant

Notice of Allowance issued in U.S. Appl. No. 18/356,587 mailed Nov. 19, 2024, 8 pages. cited by applicant

---

Primary Examiner: Gilles; Jude Jean

Attorney, Agent or Firm: JAFFERY WATSON HAMILTON & DESANCTIS LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims the benefit of and priority to U.S. application Ser. No. 17/445,163, entitled TECHNOLOGIES FOR PROTOCOL EXECUTION WITH AGGREGATION AND CACHING, by Matthias Schunter, filed Aug. 16, 2021, now allowed, which is a continuation of and claims the benefit of and priority to U.S. application Ser. No. 16/863,169, entitled TECHNOLOGIES FOR PROTOCOL EXECUTION WITH AGGREGATION AND CACHING, by Matthias Schunter, filed Apr. 30, 2020, now issued as U.S. Pat. No. 11,121,958, which is a continuation of and claims the benefit of and priority to U.S. application Ser. No. 15/860,301, entitled TECHNOLOGIES FOR PROTOCOL EXECUTION WITH AGGREGATION AND CACHING, by Matthias Schunter, filed Jan. 2, 2018, now issued as U.S. Pat. No. 10,644,984, which is a continuation of and claims the benefit of and priority to U.S. application Ser. No. 14/580,758, entitled TECHNOLOGIES FOR PROTOCOL EXECUTION WITH AGGREGATION AND CACHING, by Matthias Schunter, filed Dec. 23, 2014, now issued as U.S. Pat. No. 9,860,153, the entire contents of which are incorporated herein by reference.

BACKGROUND

(1) Central command and control devices (e.g., cloud computing devices) are often used to perform large-scale device management functions. For example, a central command device may handle the secure distribution of updates, secure key agreement management, and/or secure life-cycle management for a large number (e.g., billions) of computing devices. Currently, management protocols between a central command device and a large number of computing devices (e.g., endpoints) may be executed primarily based on a couple different approaches. One common approach to executing management protocols is to execute the management protocol with each computing device separately, which requires a large amount of resources and significant expense. Another common approach is to use a tree-based solution in which a hierarchy of management agents proxy the protocol; however, such solutions require the intermediary nodes to be trusted.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The concepts described herein are illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. Where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.
(2) FIG. **1** is a simplified block diagram of at least one embodiment of a system for protocol execution with aggregation and caching;
(3) FIG. **2** is a simplified block diagram of at least one embodiment of a command device of the system of FIG. **1**;
(4) FIG. **3** is a simplified block diagram of at least one embodiment of an environment of the command device of the system of FIG. **1**:
(5) FIG. **4** is a simplified block diagram of at least one embodiment of an environment of an aggregation device of the system of FIG. **1**;
(6) FIG. **5** is a simplified block diagram of at least one embodiment of an environment of a computing device of the system of FIG. **1**;
(7) FIG. **6** is a simplified flow diagram of at least one embodiment of a method for protocol execution that may be executed by the command device of the system of FIG. **1**;
(8) FIG. **7** is a simplified data flow diagram of at least one embodiment of a method for protocol execution that may be executed by an aggregation system of the system of FIG. **1**;
(9) FIG. **8** is a simplified flow diagram of at least one embodiment of a method for protocol

execution that may be executed by a computing device of the system of FIG. **1**; and
(10) FIG. **9** is a simplified data flow diagram of at least one embodiment of the methods of FIG. **6-8**.

DETAILED DESCRIPTION OF THE DRAWINGS

(11) While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

(12) References in the specification to “one embodiment.” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A. B, and C” can mean (A); (B); (C): (A and B); (B and C); (A and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C).

(13) The disclosed embodiments may be implemented, in some cases, in hardware, firmware, software, or any combination thereof. The disclosed embodiments may also be implemented as instructions carried by or stored on one or more transitory or non-transitory machine-readable (e.g., computer-readable) storage medium, which may be read and executed by one or more processors. A machine-readable storage medium may be embodied as any storage device, mechanism, or other physical structure for storing or transmitting information in a form readable by a machine (e.g., a volatile or non-volatile memory, a media disc, or other media device).

(14) In the drawings, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

(15) Referring now to FIG. **1**, a system **100** for protocol execution with aggregation and caching includes a command device **102**, a network **104**, one or more aggregation systems **106**, one or more computing devices **108**, and a cache store **110**. Additionally, as shown in the illustrative embodiment, each of the aggregation systems **106** may include one or more aggregation devices **112**.

(16) As described in detail below, in the illustrative embodiment, the command device **102** may handle the secure distribution of updates to computing devices **108**, the establishment of secure key agreements (e.g., between the command device **102** and the computing devices **108**), and/or perform other management-related functions. For example, the command device **102** may serve as a central command and control center (e.g., in a cloud network) and/or may be configured to execute various management protocols between the command device **102** and the computing devices **108**. In doing so, the command device **102** may broadcast messages (e.g., protocol messages) to the computing devices **108**, receive status messages and/or other response messages, and retrieve data stored in the cache store **110** by the computing devices **108** (e.g., to complete or further execution of a protocol between the command device **102** and the computing devices **108**).

(17) In some embodiments, the command device **102** may broadcast protocol messages to one or more aggregation systems **106** and utilize the aggregation systems **106** to further broadcast the protocol messages to the computing devices **108** (e.g., using a tree-based or other hierarchical approach). For example, in a tree-based approach, the command device **102** may broadcast a message to a high-level (parent) aggregation device **112** of an aggregation system **106**, which further broadcasts the message to its children aggregation devices **112**. The children aggregation devices **112** further broadcast the message to their children (i.e., the grandchildren devices of the high-level aggregation device **112**), and so on. The low-level aggregation devices **112** then transmit the message to the computing devices **108** (the endpoint nodes). Of course, in other embodiments, the system **100** and/or the aggregation systems **106** may utilize a different scheme to broadcast information (e.g., protocol messages) from the command device **102** to the computing device **108**.
(18) The computing devices **108** may then execute an operation of the protocol based on the received protocol message and store the response to the cache store **110**. As described below, it should be appreciated that the particular operation executed and the response stored may vary significantly depending on the particular protocol being executed between the command device **102** and the corresponding computing device **108**. Additionally, the cache store **110** may be embodied as any device, component, and/or structure configured to store data received from a computing device **108** and accessible to the command device **102**. For example, in some embodiments, the computing device **108** may publish the response to a storage location associated with a particular Uniform Resource Locator (URL) known to the command device **102**. Although the cache store **110** is shown in FIG. **1** as a single independent component, in some embodiments, the cache store **110** may be embodied as multiple components and/or form a portion of one or more of the aggregation systems **106** or computing devices **108**. For example, in some embodiments, a computing device **108** may publish its protocol responses to a cache store **110** located on memory or data storage of the computing device **108** itself. In other embodiments, the aggregation devices **112** and/or separate caching servers (not shown) may be utilized to store the protocol responses.
(19) In response to execution of the protocol operation(s) by the computing devices **108**, the corresponding aggregation systems **106** may receive status messages from the computing devices **108** that indicate whether the operation was a success or failure. In the illustrative embodiment, each aggregation system **106** aggregates the status messages to generate an aggregated status message for transmission to the command device **102**. As described below, the aggregated status message uniquely identifies the computing devices **108** that failed to execute the protocol operation but identify the success of the remaining computing devices **108** with a single identifier (see FIG. **7**). In other words, the aggregation system **106** may aggregate responses incoming from the computing devices **108** (e.g., by sorting and removing duplicates). In particular, each aggregation device **112** may aggregate the status responses received from its children or lower-level computing devices **108** and/or the intermediately aggregated status responses received from its children aggregation devices **112**. That is, each of the aggregation systems **106** may include one or more aggregation devices **112**, which may work cooperatively with one another to broadcast protocol messages to a corresponding set of computing devices **108** and/or aggregate status messages received from the computing devices **108** and/or “lower-level” aggregation devices **112** into an aggregated status message.
(20) Based on the received aggregated status message(s), the command device **102** may determine which computing devices **108** successfully performed the corresponding protocol operation. Additionally, the command device **102** may retrieve the protocol response message published by a particular computing device **108** to the cache store **110** at a point in time at which the command device **102** is ready to continue execution of the protocol. As such, the command device **102** may execute a protocol with many computing devices **108** (e.g., millions of devices) and handle other tasks (e.g., critical tasks for other computing devices **108**) until the command device **102** determines it is necessary or prudent to complete/continue execution of the protocol with a

particular computing device **108**. At that point, the command device **102** may retrieve the protocol response and complete/continue the protocol accordingly. In such a way, it is not necessary for the command device **102** to expend the time and/or resources to complete the execution of the protocol with the computing devices **108** in advance.

(21) The command device **102**, each of the computing devices **108**, and each of the aggregation devices **112** may be embodied as any type of computing device capable of performing the functions described herein. For example, each of the devices **102**, **108**, **112** may be embodied as a desktop computer, server, router, switch, laptop computer, tablet computer, notebook, netbook, Ultrabook™, cellular phone, smartphone, wearable computing device, personal digital assistant, mobile Internet device, Hybrid device, and/or any other computing/communication device. Although only one command device **102**, one network **104**, and one cache store **110** are illustratively shown in FIG. **1**, the system **100** may include any number of command devices **102**, networks **104**, and/or cache stores **110** in other embodiments. For example, in some embodiments, the aggregation systems **106** may include or more networks **104** (e.g., for communication between the aggregation devices **112** of the corresponding aggregation system **106**). Further, as described herein, the cache store **110** may be distributed across multiple devices in some embodiments.

(22) Referring now to FIG. **2**, an illustrative embodiment of the command device **102** is shown. As shown, the illustrative command device includes a processor **210**, an input/output ("I/O") subsystem **212**, a memory **214**, a data storage **216**, a communication circuitry **218**, and one or more peripheral devices **220**. Of course, the command device **102** may include other or additional components, such as those commonly found in a typical computing device (e.g., various input/output devices and/or other components), in other embodiments. Additionally, in some embodiments, one or more of the illustrative components may be incorporated in, or otherwise form a portion of, another component. For example, the memory **214**, or portions thereof, may be incorporated in the processor **210** in some embodiments.

(23) The processor **210** may be embodied as any type of processor capable of performing the functions described herein. For example, the processor **210** may be embodied as a single or multi-core processor(s), digital signal processor, microcontroller, or other processor or processing/controlling circuit. Similarly, the memory **214** may be embodied as any type of volatile or non-volatile memory or data storage capable of performing the functions described herein. In operation, the memory **214** may store various data and software used during operation of the command device **102** such as operating systems, applications, programs, libraries, and drivers. The memory **214** is communicatively coupled to the processor **210** via the I/O subsystem **212**, which may be embodied as circuitry and/or components to facilitate input/output operations with the processor **210**, the memory **214**, and other components of the command device **102**. For example, the I/O subsystem **212** may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.) and/or other components and subsystems to facilitate the input/output operations. In some embodiments, the I/O subsystem **212** may form a portion of a system-on-a-chip (SoC) and be incorporated, along with the processor **210**, the memory **214**, and other components of the command device **102**, on a single integrated circuit chip.

(24) The data storage **216** may be embodied as any type of device or devices configured for short-term or long-term storage of data such as, for example, memory devices and circuits, memory cards, hard disk drives, solid-state drives, or other data storage devices. The data storage **216** and/or the memory **214** may store various data during operation of the command device **102** useful for performing the functions described herein.

(25) The communication circuitry **218** may be embodied as any communication circuit, device, or collection thereof, capable of enabling communications between the command device **102** and other remote devices (e.g., the aggregation devices **112** of the aggregation systems **106**). The

communication circuitry **218** may be configured to use any one or more communication technologies (e.g., wireless or wired communications) and associated protocols (e.g., Ethernet, Bluetooth®, Wi-Fi®, WiMAX, etc.) to effect such communication.

(26) The peripheral devices **220** may include any number of additional peripheral or interface devices, such as speakers, microphones, additional storage devices, and so forth. The particular devices included in the peripheral devices **220** may depend on, for example, the type and/or intended use of the command device **102**. For example, in some embodiments, the command device **102** may be embodied as a server that has no peripheral devices **220**.

(27) Referring back to FIG. **1**, the network **104** may be embodied as any type of communication network capable of facilitating communication between the command device **102** and remote devices (e.g., the aggregation devices **112** of the aggregation systems **106**). As such, the network **104** may include one or more networks, routers, switches, computers, and/or other intervening devices. For example, the network **104** may be embodied as or otherwise include one or more cellular networks, telephone networks, local or wide area networks, publicly available global networks (e.g., the Internet), an ad hoc network, or any combination thereof.

(28) As indicated above, each of the computing devices **108** and each of the aggregation devices **112** may be embodied as any server or computing device capable of performing the functions described herein. For example, the devices **108**, **112** may similar to the command device **102**. In particular, the computing devices **108** and/or the aggregation devices **112** may include components similar to the components of the command device **102** described above and/or components commonly found in a computing device such as a processor, memory, I/O subsystem, data storage, peripheral devices, and so forth, which are not illustrated in FIG. **1** for clarity of the description.

(29) Referring now to FIG. **3**, in use, the command device **102** establishes an environment **300** for protocol execution. The illustrative environment **300** includes a protocol management module **302** and a communication module **304**. Additionally, the protocol management module **302** includes a broadcast module **306** and a data retrieval module **308** and, in some embodiments, may include a cryptography module **310**. The various modules of the environment **300** may be embodied as hardware, software, firmware, or a combination thereof. For example, the various modules, logic, and other components of the environment **300** may form a portion of, or otherwise be established by, the processor **210** or other hardware components of the command device **102**. As such, in some embodiments, one or more of the modules of the environment **300** may be embodied as a circuit or collection of electrical devices (e.g., a protocol management circuit, a communication circuit, a broadcast circuit, a data retrieval circuit, and/or a cryptography circuit). Additionally, in some embodiments, one or more of the illustrative modules may form a portion of another module and/or one or more of the illustrative modules may be embodied as a standalone or independent module.

(30) The protocol management module **302** is configured to perform various functions associated with the execution of a protocol between the command device **102** and other devices (e.g., the computing devices **108**). It should be appreciated that the particular functions performed by the protocol management module **302** may vary depending on the particular protocol. For example, a simple protocol may involve the transmission of a message to a computing device **108** (e.g., via one or more aggregation systems **106**) and a response of the computing device **108** to the command device **102** based on execution of an operation associated with the protocol or, more particularly, the transmitted message. As described above, a protocol status message may be transmitted to the command device **102**, which may be aggregated with the status messages of other computing devices **108** depending on the particular circumstances (e.g., depending on whether the operation was successful or a failure). Additionally, the computing device **108** may publish the actual response to the simple protocol message to the cache store **110** (e.g., a known URL). If the protocol is a Diffie-Hellmann Key Exchange, for example, the computing device **108** may receive the Diffie-Hellmann public key of the command device **102** (e.g., in a broadcasted message), generate the shared Diffie-Hellmann key based on its private Diffie-Hellman key, and publish its Diffie-

Hellmann public key to the cache store **110** for subsequent access by the command device **108** (i.e., for subsequent generation of the shared Diffie-Hellmann key). Of course, the protocol management module **302** may perform various other functions in order to manage the execution of other protocols in other embodiments. In other words, in some embodiments, the protocol management module **302** may perform protocol-specific functions (e.g., for cryptographic key exchanges, broadcast encryption, etc.). Additionally, in some embodiments, the protocol management module **302** may authorize a gateway device to manage or execute a protocol on behalf of the command device **102** as described below.

(31) As discussed above, the illustrative protocol management module **302** includes a broadcast module **306** and a data retrieval module **308** and, in some embodiments, may include a cryptography module **310**. The broadcast module **306** handles the broadcasting of protocol messages to various computing devices **108** (e.g., via the communication module **304**). As such, in the illustrative embodiment, the broadcast module **306** may determine the interrelationships of the various devices of the system **100**. In particular, the broadcast module **306** may determine the hierarchical relationships of the aggregation system(s) **106** (e.g., a tree-based hierarchy) and/or otherwise determine the aggregation devices **112** to which to transmit a particular protocol message in order to ensure that it is received by one or more particular computing devices **108**. In some embodiments, the command device **102** may intend to broadcast a particular protocol message to all computing devices **108** in the system **100** in which case the broadcast module **306** may transmit the protocol message to each of the aggregation systems **106** for further dissemination. In other embodiments, the broadcast module **306** may identify a subset of computing devices **108** of the system **100** that should receive a particular protocol message and transmit the message to the appropriate aggregation system(s) **106** or directly to the computing device(s) **108** accordingly.

(32) The data retrieval module **308** handles the retrieval of data from the cache store **110**. As discussed above, the computing devices **108** may publish (e.g., to a URL) or otherwise store responses to protocol messages received from the command device **102** (e.g., via the aggregation systems **106**). As such, the data retrieval module **308** may determine the storage location within the cache store **110** at which to retrieve a response to a particular protocol message. In some embodiments, the command device **102** and the computing devices **108** utilize a specific naming and/or storage scheme so that the computing devices **108** store response data at locations from which the command device **102** expects to retrieve the data. In other embodiments, the broadcast module **306** of the command device **102** may identify the location at which the computing device **108** is to store/publish the response data in the broadcasted protocol message. It should be appreciated that, in some embodiments, multiple protocol messages between the command device **102** and a particular computing device **108** (e.g., “rounds”) may be independent of one another such that the computing device **108** may perform operations associated with a “second” protocol message without having performed operations associated with a “first” protocol message. As such, the protocol may be decomposed into a sequence of protocol executions/rounds that may be broadcasted to the computing device **108** without intervening input from the command device **102**. In those embodiments, the computing device **108** may publish/store the responses to the multiple rounds at the same storage location of the cache store **110** or at different (e.g., uniquely identifiable) storage locations of the cache store **110**. Further, in some embodiments, the computing device **108** may store a failure log and/or other information associated with protocol execution in the cache store **110** and accessible to the command device **102**.

(33) The cryptography module **310** may perform various security-related functions (e.g., attestation and cryptography) for the command device **102**. As indicated above, the particular cryptographic functions performed by the cryptography module **310** may vary depending on the particular embodiment (e.g., depending on the particular protocol executed by the command device **102**). In various embodiments, the cryptography module **310** may perform encryption/decryption, cryptographic signatures, cryptographic key generation (e.g., asymmetric and/or symmetric key

generation), cryptographic hash generation, and/or other cryptographic functions. Further, in some embodiments, the cryptography module **310** may be configured to establish a trusted execution environment. For example, in some embodiments, the cryptography module **310** may be embodied as a security co-processor, such as a trusted platform module (TPM), a secure enclave such as Intel® Software Guard Extensions (SGX), or an out-of-band processor. Additionally, in some embodiments, the cryptography module **310** may establish an out-of-band communication link with remote devices.

(34) The communication module **304** handles the communication between the command device **102** and remote computing devices (e.g., the aggregation systems **106**, the computing devices **108**, the cache store **110**, etc.) through the network **104**. For example, as described herein, the communication module **304** may broadcast protocol messages to the computing devices **108** (e.g., through the aggregation systems **106**) and retrieve responses of the computing devices **108** from the cache store **110**.

(35) Referring now to FIG. **4**, in use, each aggregation device **112** establishes an environment **400** for protocol execution. The illustrative environment **400** includes a broadcast module **402**, an aggregation module **404**, and a communication module **406**. The various modules of the environment **400** may be embodied as hardware, software, firmware, or a combination thereof. For example, the various modules, logic, and other components of the environment **400** may form a portion of, or otherwise be established by, the processor or other hardware components of the aggregation device **112**. As such, in some embodiments, one or more of the modules of the environment **400** may be embodied as a circuit or collection of electrical devices (e.g., a broadcast circuit, an aggregation circuit, and/or a communication circuit). Additionally, in some embodiments, one or more of the illustrative modules may form a portion of another module (e.g., the broadcast module **402** may form a portion of the communication module **406**).

(36) The broadcast module **402** handles the broadcasting of protocol messages received from the command device **102** and/or a higher-level aggregation device **112** to lower-level aggregation devices **112** and/or computing devices **108** (e.g., via the communication module **406**). As described above, the aggregation device **112** may form a portion of a particular aggregation system **106** in some hierarchical arrangement relative to the other aggregation devices **112** of the aggregation system **106** (e.g., in a tree-based hierarchy). Accordingly, in the illustrative embodiment, the broadcast module **402** transmits received protocol messages to other aggregation devices **112** further "down" the hierarchy or directly to the computing device(s) **108** depending on the position of the aggregation device **112** in the hierarchy. In some embodiments, the received protocol messages may have particular recipients in which case the broadcast module **402** may determine its hierarchical relationship relative to other devices in order to transmit the protocol messages to the appropriate destination(s). For example, the broadcast module **402** may determine its hierarchical relationship relative to other aggregation devices **112** within the same aggregation system **106**, relative to other aggregation systems **106** or the devices **112** therein, relative to the command device **102**, and/or relative to the computing devices **108**.

(37) The aggregation module **404** is configured to aggregate status messages associated with the success/failure of protocol execution by the computing devices **108**. In particular, a computing device **108** may attempt to perform operations (e.g., execute particular instructions) associated with a protocol message received from the command device **102** (e.g., through the aggregation systems **106**) and generate a status message that indicates whether the operations were performed successfully. As discussed above, in the illustrative embodiment, the aggregation device **112** occupies a particular hierarchical position in the corresponding aggregation system **106**. Depending on the particular embodiment (e.g., the particular hierarchical position), the aggregation module **404** of the aggregation device **112** aggregates the status messages received from the corresponding computing devices **108** and/or lower level aggregation devices **112** (e.g., within the same aggregation system **106**) into a single aggregated status message. For clarity, the aggregated status

messages generated by aggregation devices **112** other than the highest-level aggregation devices **112** of an aggregation system **106** (or of the system **100** generally) may be described herein as intermediate aggregated status messages. As described herein, in aggregating the lower-level status messages, the aggregation module **404** removes duplicate status messages and/or other data duplication. For example, in some embodiments, the aggregation module **404** generates an aggregated status message based on the received status messages (or intermediate aggregated status messages) in which the aggregated status message uniquely identifies the computing devices **108** that failed and identifies the success of the remaining computing devices **108** with a single success identifier (see FIG. **7**).

(38) The communication module **406** handles the communication between the aggregation device **112** and remote computing devices (e.g., the command device **102**, other aggregation devices **112**, etc.) through one or more networks (e.g., the network **104**). For example, as described herein, the communication module **406** may broadcast protocol messages to lower-level aggregation devices **112** and/or the computing devices **108** and transmit aggregated status messages to the command device **102** and/or higher-level aggregation devices **112**.

(39) Referring now to FIG. **5**, in use, each computing device **108** establishes an environment **500** for protocol execution. The illustrative environment **500** includes a protocol execution module **502** and a communication module **504**. Additionally, the protocol execution module **502** includes a publication module **506** and a status module **508** and, in some embodiments, may include a cryptography module **510**. The various modules of the environment **500** may be embodied as hardware, software, firmware, or a combination thereof. For example, the various modules, logic, and other components of the environment **500** may form a portion of, or otherwise be established by, the processor or other hardware components of the computing device **108**. As such, in some embodiments, one or more of the modules of the environment **500** may be embodied as a circuit or collection of electrical devices (e.g., a protocol execution circuit, a communication circuit, a publication circuit, a status circuit, and/or a cryptography circuit). Additionally, in some embodiments, one or more of the illustrative modules may form a portion of another module and/or one or more of the illustrative modules may be embodied as a standalone or independent module.

(40) The protocol execution module **502** is configured to perform various functions associated with the execution of a protocol between the computing device **108** and other devices (e.g., the command device **102**). Similar to the protocol management module **302** described above, the particular functions performed by the protocol execution module **502** of the computing device **108** may vary depending on the particular protocol. For example, as described above, some protocols may involve the generation and/or exchange of cryptographic keys, whereas other protocols may require the performance of other functions by the protocol execution module **502**.

(41) As discussed above, the illustrative protocol execution module **502** includes a publication module **506** and a status module **508** and, in some embodiments, may include a cryptography module **510**. The publication module **506** publishes and/or otherwise stores responses to protocol messages received from the command device **102** (e.g., via the aggregation systems **106**). In particular, in the illustrative embodiment, the publication module **506** stores responses to protocol messages in the cache store **110** for subsequent access by the command device **102**. For example, as described above, the publication module **506** may publish the response to a storage location associated with a particular URL known to the command device **102**. In some embodiments, the storage location may be predetermined by the computing device **108** and/or the command device **102**, for example, according to an understood naming and storage scheme. Further, as described above, the command device **102** may broadcast a storage location at which the computing device **108** is to store a particular protocol response in some embodiments. Additionally, in some embodiments, the publication module **506** may store a protocol execution log that describes the protocol operations performed by the computing device **108** and/or results of protocol operations (e.g., a failure log). In some embodiments, the command device **102** may subsequently access the

protocol execution log (e.g., to remedy failed execution of a protocol operation).
(42) The status module **508** transmits status messages regarding the status of execution of one or more operations associated with a protocol message received from the command device **102**. For example, in the illustrative embodiment, the status module **508** determines whether a protocol operation corresponding with a protocol message was successful, generates a status response message that indicates the success or failure of the protocol operation, and transmits the status response message to the command device **102**. As described above, in the illustrative embodiment, the computing device **108** transmits the status message to the corresponding aggregation system **106** of the computing device **108**, which aggregates the status message of the computing device **108** with status messages of other computing devices **108** to generate an aggregated status message for the command device **102**.
(43) The cryptography module **510** may perform various security-related functions (e.g., attestation and cryptography) for the computing device **108**. In some embodiments, the cryptography module **510** may be similar to the cryptography module **310** of the command device **102** described above. As such, in various embodiments, the cryptography module **510** may perform encryption/decryption, cryptographic signatures, cryptographic key generation (e.g., asymmetric and/or symmetric key generation), cryptographic hash generation, and/or other cryptographic functions. Further, in some embodiments, the cryptography module **510** may be configured to establish a trusted execution environment, and/or may be embodied as a security co-processor, such as a trusted platform module (TPM), a secure enclave such as Intel® Software Guard Extensions (SGX), or an out-of-band processor. Additionally, in some embodiments, the cryptography module **510** may establish an out-of-band communication link with remote devices.
(44) The communication module **504** handles the communication between the computing device **108** and remote computing devices (e.g., the aggregation systems **106**, the cache store **110**, the command device **102**, etc.) through the network **104**. For example, as described herein, the communication module **504** may receive broadcasted protocol message from the command device **102** (e.g., through the aggregation systems **106**), publish protocol message responses to the cache store **110**, and/or transmit status messages to the command device **102** (e.g., in aggregated form via the aggregation systems **106**).
(45) Referring now to FIG. **6**, in use, the command device **102** may execute a method **600** for protocol execution. The illustrative method **600** begins with block **602** in which the command device **102** broadcasts a protocol message to one or more computing devices **108**. As described above, in some embodiments, the system **100** may include one or more intervening aggregation systems **106** that may cooperate with one another to broadcast protocol messages from the command device **102** to a plurality of computing devices **108**. In doing so, it should be appreciated that the aggregation systems **106** may reduce the network and/or resource load on the command device **102** associated with communicating with each of the computing devices **108** individually. As such, in block **604**, the command device **102** transmits the protocol message to the aggregation system(s) **106** (see data flow **902** of method **900** of FIG. **9**) for further broadcasting to the computing devices **108** (see data flow **904** of method **900** of FIG. **9**). In particular, in the illustrative embodiment, the command device **102** transmits the protocol message to the highest-level aggregation device(s) **112** in each of the corresponding aggregation systems **106**. As discussed above, in some embodiments, the protocol message is directed to only a subset of computing devices **108** of the system **100**, in which case the command device **102** may broadcast the protocol message to the appropriate aggregation systems **106** and notify those aggregation systems **106** of the destination computing devices **108**.
(46) As described herein, in the illustrative embodiment, each of the recipient computing devices **108** may perform a protocol operation associated with the received protocol message, publish a response to the cache store **110**, and transmit a status message to the aggregation system(s) **106**. Further, in the illustrative embodiment, the aggregation system(s) **106** aggregate the status

messages of multiple computing devices **108** into one or more aggregated status messages as described above. Accordingly, the command device **102** receives the aggregated status message(s) from the aggregation system(s) **106** in block **606** (see data flow **912** of method **900** of FIG. **9**).

(47) In block **608**, the command device **102** determines whether to continue execution of the protocol (i.e., the protocol associated with the broadcasted protocol message) with a particular computing device **108**. If so, in block **610**, the command device **102** retrieves the computing device **108**'s response to the protocol message from the cache store **110**. As discussed above, the storage location of the particular response in the cache store **110** is known and accessible to the command device **102**. For example, in block **612**, the command device **102** may retrieve the protocol message response from a known URL to which the computing device **108** published the response. It should be appreciated that, in some embodiments, a significant amount of time may lapse between receipt of the aggregated status message by the command device **102** and retrieval of the protocol message response by the command device **102**. As such, in some embodiments, the command device **102** may determine to continue protocol execution with a particular computing device **108** at a convenient time depending on the particular protocol. For example, for a cryptographic key exchange protocol, the command device **102** may retrieve a cryptographic key of the computing device **108** from the cache store **110** when the cryptographic key is needed by the command device **102** (e.g., to securely communicate with the computing device **108**).

(48) Referring now to FIG. **7**, in use, the aggregation system **106** may execute a method **700** for protocol execution. It should be appreciated that the method **700** depicts data flows that may be employed in a tree-based hierarchy to aggregate the status messages of the corresponding computing devices **108**. As shown in FIG. **7**, the method **700** depicts four computing devices **108** (D1, D2, D3, and D4), an aggregation system **106** including three aggregation devices **112**, and the command device **102**. It should be appreciated that those quantities of devices are depicted in FIG. **7** for simplicity and clarity of the description. However, in other embodiments, the system **100** may include any number of computing devices **108**, aggregation systems **106**, and aggregation devices **112** as described above in which similar principles may be employed. Further, the aggregation systems **106** and/or the aggregation devices **112** may be arranged in any suitable hierarchical relationship/structure.

(49) As shown in the illustrative embodiment of FIG. **7**, the aggregation system **106** receives status messages from each of the four computing devices **108** (see data flow **908** of method **900** of FIG. **9**). In particular, a first aggregation device **112** receives status messages from the computing devices **108** D1 and D2 that indicate success (OK) and failure (FAIL(D2)) of the corresponding protocol operations, respectively. Further, a second aggregation device **112** receives status messages from the computing devices **108** D3 and D4 that indicate success (OK) and failure (FAIL(D4)) of the corresponding protocol operations, respectively. As shown, the first aggregation device **112** aggregates the status messages of the computing devices **108** D1 and D2 into a single aggregated status message (OK, FAIL(D2)) and transmits that aggregated status message to the third aggregation device **112**. Similarly, the second aggregation device **112** aggregates the status messages of the computing devices **108** D3 and D4 into a single aggregated status message (OK, FAIL(D4)) and transmits that aggregated status message to the third aggregation device **112**. In the illustrative embodiment, the third aggregation device **112** aggregates the intermediate aggregated status messages received from the first and second aggregation devices **112** into another aggregated status message (OK, FAIL(D2), FAIL(D4)), which is transmitted to the command device **102**. As described above, in the illustrative embodiment, the aggregation devices **112** of the aggregation system **106** aggregate the status messages by removing duplicate status messages that identify protocol execution success and maintaining unique identifiers of computing devices **108** that failed to execute the protocol (see data flow **910** of method **900** of FIG. **9**).

(50) It should be appreciated that the aggregated status messages generated by the first and second aggregation devices **112** may be described herein as an intermediate aggregation status message in

order to distinguish the aggregated status message from the aggregated status message generated by the third aggregation device **112**, which is transmitted to the command device **102**. Additionally, in more complex hierarchies, it should be appreciated that a particular aggregation device **112** may receive status messages from any combination of other aggregation devices **112** and/or computing devices **108**.

(51) Referring now to FIG. **8**, in use, one or more of the computing devices **108** may execute a method **800** for protocol execution. The illustrative method **800** begins with block **802** in which the computing device **108** receives a protocol message from the corresponding aggregation system **106**. As described above, in the illustrative embodiment, the command device **102** utilizes the aggregation system **106** to broadcast the protocol message to the computing device **108** and the protocol message is associated with the execution of a protocol between the command device **102** and the computing device **108**. As such, the computing device **108** may perform (or attempt to perform) the corresponding protocol operation based on the protocol message and publish a response to the protocol message to the cache store **110** in block **804** (see data flow **906** of method **900** of FIG. **9**). For example, in block **806**, the computing device **108** may publish the response to a URL known the command device **102** as described above. Of course, in some embodiments, the computing device **108** may be unable to execute the particular protocol operation in which case the computing device **108** may, for example, publish a failure log to the cache store **110** instead. In block **808**, the computing device **108** transmits a status message that indicates the success/failure of the protocol execution to the corresponding aggregation system **106**.

(52) It should be appreciated that the techniques described herein may be employed by the system **100** for aggregation and protocol response caching during protocol execution for any number of protocols executed between the command device **102** and the computing devices **108**. For example, as described above, the techniques may be utilized during the execution of a Diffie-Hellmann key exchange between the command device **102** and the computing devices **108**. In such an embodiment, the command device **102** may select a prime number, p, and determine a primate root or generator, g (mod p). Further, the command device **102** may select another integer, a, as its private Diffie-Hellmann key and calculate its public Diffie-Hellmann key as A=g.sup.a mod p. The command device **102** broadcasts the public Diffie-Hellmann key, A, as well as the generator, g, and the prime number, p, to one or more of the computing devices **108** (e.g., through the aggregation system(s) **106**). One of the computing devices **108** may then select an integer, b, as its own private Diffie-Hellmann key and calculate its public Diffie-Hellmann key as B=g.sup.b mod p. Further, the computing device **108** may calculate the shared Diffie-Hellmann key (A.sup.b=g.sup.ab) for secure communication and store that key. Additionally, the computing device **108** may publish its public Diffie-Hellmann key, B, to the cache store **110** (e.g., a known URL) and transmit a status message to the aggregation system **106** indicating that the protocol operation was successfully executed (“OK”). As such, the command device **102** may subsequently retrieve the public Diffie-Hellman key of the computing device **108** from the cache store **110**, calculate the shared Diffie-Hellmann key (B.sup.a=g.sup.ab), and securely communicate with the computing device **108** using the shared cryptographic key. Of course, as described above, if the computing device **108** is unable to successfully perform the relevant protocol operations, the computing device **108** may instead publish a failure log to the cache store **110** and indicate that protocol execution was a failure in the protocol status message.

(53) In another embodiment, the command device **102** may utilize the techniques described herein to efficiently establish cryptographic keys for a broadcast encryption scheme. It should be appreciated that in a broadcast encryption scheme, the cryptographic keys may be structured in a hierarchy to allow broadcasting to subsets of entities (e.g., associated with particular cryptographic keys). In some embodiments, the system **100** may be structured as a tree-based hierarchy in which each entity of the system **100** may be considered to be a node such that the command device **102** (Node N) is the highest-level node, the next lower-level nodes are identified as Nodes N.sub.0,

N.sub.1, the next lower-level nodes are identified as Nodes N.sub.00, N.sub.01, N.sub.10, and N.sub.11, and so on (N.sub.00 and N.sub.01 being children of N.sub.0 and N.sub.10 and N.sub.11 being children of N.sub.1). In such embodiments, each node N.sub.xy generates a cryptographic key pair PK(N.sub.xy) and SK(N.sub.xy). Further, PK(N.sub.xy) is broadcasted to the children of the node N.sub.xy and published to the cache store **110** (e.g., a particular URL) and associated with N.sub.xy (e.g., URL(N.sub.xy)). It should be appreciated that the goal of such key distribution is to ensure that each child node has the public keys of its ancestors. In such embodiments, in order to broadcast an encrypted message to all of the nodes, for example, the command device **102** (i.e., node N) or another computing device may retrieve PK(N) from the cache store **110** (e.g., at URL(N)), encrypt the message under PK(N), and broadcast the encrypted message to all of the nodes. It should be appreciated that, in some embodiments, one or more cryptographic keys may be revoked and the keys on the "path" to the revoked nodes may be revoked. Instead, the highest remaining cryptographic keys may be utilized to broadcast messages. For example, when the cryptographic key for N.sub.10 is revoked, the keys PK(N.sub.0) and PK(N.sub.11) may be used to encrypt future broadcasted messages.

(54) In some embodiments, it should be appreciated that the command device **102** may delegate authority to execute a protocol between the command device **102** and one or more computing devices **108** to a gateway device. In such embodiments, the gateway device may communicate with a particular computing device **108** to execute one or more rounds of a protocol and store the results of the protocol execution for subsequent access by the command device **102**. For example, the gateway device may generate a protocol transcript identifying the functions performed, communications transmitted/received, results, and/or other information associated with the delegated protocol execution. Further, in some embodiments, the gateway device and/or the corresponding computing device **108** may publish the transcript to the cache store **110** at a storage location known and accessible to the command device **102**.

EXAMPLES

(55) Illustrative examples of the technologies disclosed herein are provided below. An embodiment of the technologies may include any one or more, and any combination of, the examples described below. Example 1 includes a command device for protocol execution, the command device comprising a broadcast module to broadcast a protocol message to a plurality of computing devices; and a communication module to receive an aggregated status message from an aggregation system, wherein the aggregated status message identifies a success or failure of execution of instructions corresponding with the protocol message by the plurality of computing devices such that each computing device of the plurality of computing devices that failed is uniquely identified and the success of remaining computing devices is aggregated into a single success identifier. Example 2 includes the subject matter of Example 1, and further including a data retrieval module to retrieve, from a cache store, a response to the protocol message of a computing device of the plurality of computing devices, wherein the cache store is to store responses of the plurality of computing devices to the protocol message for subsequent access to the responses by the command device. Example 3 includes the subject matter of any of Examples 1 and 2, and wherein to retrieve the response from the cache store comprises to retrieve the response from a cache store accessed at a uniform resource locator known to both the command device and the corresponding computing device. Example 4 includes the subject matter of any of Examples 1-3, and wherein to broadcast the protocol message comprises to identify a storage location of the cache store at which each computing device of the plurality of computing devices is to store a response to the protocol message. Example 5 includes the subject matter of any of Examples 1-4, and wherein to broadcast the protocol message comprises to broadcast a first protocol message of a multi-round protocol; and wherein the broadcast module is further to broadcast a second protocol message of the multi-round protocol to the plurality of computing devices prior to retrieval of the response to the first protocol message. Example 6 includes the subject matter of any of Examples 1-5, and further

including a data retrieval module to retrieve, from a cache store, a failure log of a computing device of the plurality of computing devices in response to an indication by the aggregated status message that the computing device failed to execute the instructions. Example 7 includes the subject matter of any of Examples 1-6, and wherein to broadcast the protocol message comprising to broadcast the protocol message to at least one aggregation device of an aggregation system to be further broadcasted to the plurality of computing devices. Example 8 includes a method for protocol execution by a command device, the method comprising broadcasting, by the command device, a protocol message to a plurality of computing devices; and receiving, by the command device, an aggregated status message from an aggregation system, wherein the aggregated status message identifies a success or failure of execution of instructions corresponding with the protocol message by the plurality of computing devices such that each computing device of the plurality of computing devices that failed is uniquely identified and the success of remaining computing devices is aggregated into a single success identifier. Example 9 includes the subject matter of Example 8, and further including retrieving, by the command device and from a cache store, a response to the protocol message of a computing device of the plurality of computing devices, wherein the cache store is to store responses of the plurality of computing devices to the protocol message for subsequent access to the responses by the command device. Example 10 includes the subject matter of any of Examples 8 and 9, and wherein retrieving the response from the cache store comprises retrieving the response from a cache store accessed at a uniform resource locator known to both the command device and the corresponding computing device. Example 11 includes the subject matter of any of Examples 8-10, and wherein broadcasting the protocol message comprises identifying a storage location of the cache store at which each computing device of the plurality of computing devices is to store a response to the protocol message. Example 12 includes the subject matter of any of Examples 8-11, and wherein broadcasting the protocol message comprises broadcasting a first protocol message of a multi-round protocol; and further comprising broadcasting a second protocol message of the multi-round protocol to the plurality of computing devices prior to retrieving the response to the first protocol message. Example 13 includes the subject matter of any of Examples 8-12, and further including retrieving, by the command device and from a cache store, a failure log of a computing device of the plurality of computing devices in response to the aggregated status message indicating that the computing device failed to execute the instructions. Example 14 includes the subject matter of any of Examples 8-13, and wherein broadcasting the protocol message comprising broadcasting the protocol message to at least one aggregation device of an aggregation system to be further broadcasted to the plurality of computing devices. Example 15 includes a computing device comprising a processor; and a memory having stored therein a plurality of instructions that when executed by the processor cause the computing device to perform the method of any of Examples 8-14. Example 16 includes one or more machine-readable storage media comprising a plurality of instructions stored thereon that, in response to execution by a computing device, cause the computing device to perform the method of any of Examples 8-14. Example 17 includes a command device for protocol execution, the command device comprising means for broadcasting a protocol message to a plurality of computing devices; and means for receiving an aggregated status message from an aggregation system, wherein the aggregated status message identifies a success or failure of execution of instructions corresponding with the protocol message by the plurality of computing devices such that each computing device of the plurality of computing devices that failed is uniquely identified and the success of remaining computing devices is aggregated into a single success identifier. Example 18 includes the subject matter of Example 17, and further including means for retrieving, from a cache store, a response to the protocol message of a computing device of the plurality of computing devices, wherein the cache store is to store responses of the plurality of computing devices to the protocol message for subsequent access to the responses by the command device. Example 19 includes the subject matter of any of Examples 17 and 18, and wherein the means for retrieving the response from the cache store comprises means for retrieving the response from a cache store accessed at a uniform resource locator known to both the command device and the corresponding computing device. Example 20 includes the subject matter of any of Examples 17-19, and wherein the means for broadcasting the protocol message comprises means for identifying a storage location of the cache store at which each computing device of the plurality of computing devices is to store a response to the protocol message. Example 21 includes the subject matter of any of Examples 17-20, and wherein the means for broadcasting the protocol message comprises means for broadcasting a first protocol message of a multi-round protocol; and further comprising means for broadcasting a second protocol message of the multi-round protocol to the plurality of computing devices prior to retrieving the response to the first protocol message. Example 22 includes the subject matter of any of Examples 17-21, and further including means for retrieving, from a cache store, a failure log of a computing device of the plurality of computing devices in response to the aggregated status message indicating that the computing device failed to execute the instructions. Example 23 includes the subject matter of any of Examples 17-22, and wherein the means for broadcasting the protocol message comprising means for broadcasting the protocol message to at least one aggregation device of an aggregation system to be further broadcasted to the plurality of computing devices. Example 24 includes a computing device for protocol execution, the computing device comprising a communication module to receive a protocol message from an aggregation system, wherein the protocol message corresponds with execution of a protocol between the computing device and a command device; and a protocol execution module to (i) publish a response to the protocol message to a cache store for subsequent access by the command device and (ii) transmit a status message to the aggregation system, wherein the status message indicates a success or failure of execution of instructions corresponding with the protocol message. Example 25 includes the subject matter of Example 24, and wherein to publish the response comprises to publish the response to a cache store accessed at a uniform resource locator known to both the command device and the computing device. Example 26 includes the subject matter of any of Examples 24 and 25, and wherein to publish the response comprises to publish a failure log to the cache store in response to a failure to execute the instructions corresponding with the protocol message. Example 27 includes the subject matter of any of Examples 24-26, and wherein the protocol execution module is further to perform a protocol operation associated with the received protocol message, wherein the published response is associated with the performed protocol operation. Example 28 includes a method for protocol execution by a computing device, the method comprising receiving, by the computing device, a protocol message from an aggregation system, wherein the protocol message corresponds with execution of a protocol between the computing device and a command device; publishing, by the computing device, a response to the protocol message to a cache store for subsequent access by the command device; and transmitting, by the computing device, a status message to the aggregation system, wherein the status message indicates a success or failure of execution of instructions corresponding with the protocol message. Example 29 includes the subject matter of Examples 28, and wherein publishing the response comprises publishing the response to a cache store accessed at a uniform resource locator known to both the command device and the computing device. Example 30 includes the subject matter of any of Examples 27 and 28, and where publishing the response comprises publishing a failure log to the cache store in response to a failure to execute the instructions corresponding with the protocol message. Example 31 includes the subject matter of any of Examples 27-29, and further including performing, by the computing device, a protocol operation associated with the received protocol message, wherein the published response is associated with the performed protocol operation. Example 32 includes a computing device comprising a processor; and a memory having stored therein a plurality of instructions that when executed by the processor cause the computing device to perform the method of any of Examples 28-31. Example 33 includes one or more machine-readable storage media comprising a plurality of instructions stored thereon that, in response to execution by a computing device, cause the computing device to perform the method of any of Examples 28-31. Example 34 includes a computing device for protocol execution, the computing device comprising means for receiving a protocol message from an aggregation system, wherein the protocol message corresponds with execution of a protocol between the computing device and a command device; means for publishing a response to the protocol message to a cache store for subsequent access by the command device; and means for transmitting a status message to the aggregation system, wherein the status message indicates a success or failure of execution of instructions corresponding with the protocol message. Example 35 includes the subject matter of Example 34, and wherein the means for publishing the response comprises means for publishing the response to a cache store accessed at a uniform resource locator known to both the command device and the computing device. Example 36 includes the subject matter of any of Examples 34 and 35, and where the means for publishing the response comprises means for publishing a failure log to the cache store in response to a failure to execute the instructions corresponding with the protocol message. Example 37 includes the subject matter of any of Examples 34-36, and further including means for performing a protocol operation associated with the received protocol message, wherein the published response is associated with the performed protocol operation. Example 38 includes an aggregation system for protocol execution, the aggregation system comprising a plurality of aggregation devices, each of the aggregation devices comprising a broadcast module to broadcast protocol messages received from a command device to a plurality of computing devices; and an aggregation module to aggregate status messages received from at least one of other aggregation devices or computing devices of the plurality of computing devices; wherein the aggregation system is to receive a status message from each computing device of the plurality of computing devices, wherein the status message indicates a success or failure of a protocol operation by the corresponding computing device that is associated with a protocol message of a protocol between the command device and the corresponding computing device; generate an aggregated status message based on the status messages received from the plurality of computing devices, wherein the aggregated status message uniquely identifies computing devices that failed to perform the protocol operation and identifies the success of remaining computing devices with a single success identifier; and transmit the aggregation status message to the command device. Example 39 includes the subject matter of Example 38, and wherein the aggregation system is further to receive, from the command device, the protocol message corresponding with execution of the protocol; and transmit the protocol message to the plurality of computing devices. Example 40 includes a method for protocol execution by an aggregation system, the method comprising receiving, by the aggregation system, a status message from each computing device of a plurality of computing devices, wherein the status message indicates a success or failure of a protocol operation by the corresponding computing device that is associated with a protocol message of a protocol between a command device and the corresponding computing device; generating, by the aggregation system, an aggregated status message based on the status messages received from the plurality of computing devices, wherein the aggregated status message uniquely identifies computing devices that failed to perform the protocol operation and identifies the success of remaining computing devices with a single success identifier; and transmitting, by the aggregation system, the aggregation status message to the command device. Example 41 includes the subject matter of Example 40, and further including receiving, by the aggregation system and from the command device, the protocol message corresponding with execution of the protocol; and transmitting, by the aggregation system, the protocol message to the plurality of computing devices. Example 42 includes one or more machine-readable storage media comprising a plurality of instructions stored thereon that, in response to execution by a computing system, cause the computing system to perform the method of any of Examples 40-41. Example 43 includes an aggregation system for protocol execution, the aggregation system comprising means for receiving a status message from each computing device of a plurality of computing devices, wherein the status message indicates a success or failure of a protocol operation by the corresponding computing device that is associated with a protocol message of a protocol between a command device and the corresponding computing device; means for generating an aggregated status message based on the status messages received from the plurality of computing devices, wherein the aggregated status message uniquely identifies computing devices that failed to perform the protocol operation and identifies the success of remaining computing devices with a single success identifier; and means for transmitting the aggregation status message to the command device. Example 44 includes the subject matter of Example 43, and further including means for receiving, from the command device, the protocol message corresponding with execution of the protocol; and means for transmitting the protocol message to the plurality of computing devices.

Claims

1. A command device comprising: processing circuitry coupled to a memory, the processing circuitry to: receive an aggregated status message that identifies successes or failures associated with execution of instructions corresponding to a protocol message associated with computing devices such that a failure associated with a computing device of the computing devices is uniquely identified, wherein one or more successes associated with one or more of the computing devices are aggregated into a single success identifier.
2. The command device of claim 1, wherein the processing circuitry is further to: broadcast the protocol message to the computing devices; and retrieve, from a memory store, responses associated with the protocol message, wherein the memory store to store the responses for subsequent access to the responses by the command devices, wherein the responses are retrieved from the memory store using a uniform resource locator accessible to the computing devices.
3. The command device of claim 2, wherein to broadcast the protocol message comprises to identify a storage location of the memory store where the responses associated with the protocol message are stored.
4. The command device of claim 2, wherein to broadcast the protocol message further comprises to broadcast a first protocol message associated with a multi-round protocol or a second protocol message associated with the multi-round protocol to the computing devices prior to retrieval of a first response associated with the first protocol message or a second response associated with the second protocol message.
5. The command device of claim 2, wherein the processing circuitry is further to retrieve, from the memory store, a failure log associated with the failure of the computing device in response to an indication by the aggregated status message that the computing device failed to execute the instructions.
6. The command device of claim 2, wherein to broadcast the protocol message further comprises to broadcast the protocol message to an aggregation device associated with an aggregation system to be further broadcasted to the computing devices.
7. The command device of claim 1, wherein the processing circuitry comprises one or more of application processing circuitry or graphics processing circuitry.
8. A method comprising: receiving, by a command computing device, an aggregated status message that identifies successes or failures associated with execution of instructions corresponding to a protocol message associated with computing devices such that a failure associated with a computing device of the computing devices is uniquely identified, wherein one or more successes associated with one or more of the computing devices are aggregated into a single success identifier.
9. The method of claim 8, further comprising: broadcasting the protocol message to the computing devices; and retrieving, from a memory store, responses associated with the protocol message, wherein the memory store to store the responses for subsequent access to the responses by the

command devices, wherein the responses are retrieved from the memory store using a uniform resource locator accessible to the computing devices.

10. The method of claim 9, wherein broadcasting the protocol message comprises identifying a storage location of the memory store where the responses associated with the protocol message are stored.

11. The method of claim 9, wherein broadcasting the protocol message further comprises broadcasting a first protocol message associated with a multi-round protocol or a second protocol message associated with the multi-round protocol to the computing devices prior to retrieval of a first response associated with the first protocol message or a second response associated with the second protocol message.

12. The method of claim 9, further comprising retrieving, from the memory store, a failure log associated with the failure of the computing device in response to an indication by the aggregated status message that the computing device failed to execute the instructions.

13. The method of claim 9, wherein broadcasting the protocol message further comprises broadcasting the protocol message to an aggregation device associated with an aggregation system to be further broadcasted to the computing devices.

14. The method of claim 8, wherein the command computing device comprises processing circuitry coupled to a memory, the processing circuitry having one or more of application processing circuitry or graphics processing circuitry.

15. At least one non-transitory computer-readable medium having stored thereon instructions which, when executed, cause a command computing device to perform operations comprising: receiving an aggregated status message that identifies successes or failures associated with execution of instructions corresponding to a protocol message associated with computing devices such that a failure associated with a computing device of the computing devices is uniquely identified, wherein one or more successes associated with one or more of the computing devices are aggregated into a single success identifier.

16. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise: broadcasting the protocol message to the computing devices; and retrieving, from a memory store, responses associated with the protocol message, wherein the memory store to store the responses for subsequent access to the responses by the command devices, wherein the responses are retrieved from the memory store using a uniform resource locator accessible to the computing devices.

17. The non-transitory computer-readable medium of claim 16, wherein broadcasting the protocol message comprises identifying a storage location of the memory store where the responses associated with the protocol message are stored.

18. The non-transitory computer-readable medium of claim 16, wherein broadcasting the protocol message further comprises broadcasting a first protocol message associated with a multi-round protocol or a second protocol message associated with the multi-round protocol to the computing devices prior to retrieval of a first response associated with the first protocol message or a second response associated with the second protocol message.

19. The non-transitory computer-readable medium of claim 16, wherein the operations further comprise retrieving, from the memory store, a failure log associated with the failure of the computing device in response to an indication by the aggregated status message that the computing device failed to execute the instructions.

20. The non-transitory computer-readable medium of claim 16, wherein broadcasting the protocol message further comprises broadcasting the protocol message to an aggregation device associated with an aggregation system to be further broadcasted to the computing devices, wherein the command computing device comprises processing circuitry coupled to a memory, the processing circuitry having one or more of application processing circuitry or graphics processing circuitry.